Walks

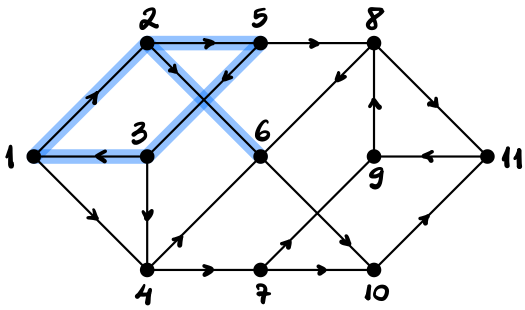
Let $D := (V, A)$ be a digraph. Let $W := \langle v_0, v_1, \ldots, v_\ell \rangle$ be a walk in D.

The walk W is a **trail** if all its ~~vertices~~ arcs are distinct,

i.e., if the function $i \in [\ell] \longmapsto v_{i-1}v_i \in A$ is injective

---

Example



$\langle 2, 5, 3, 1, 2, 6 \rangle$ is a **trail**

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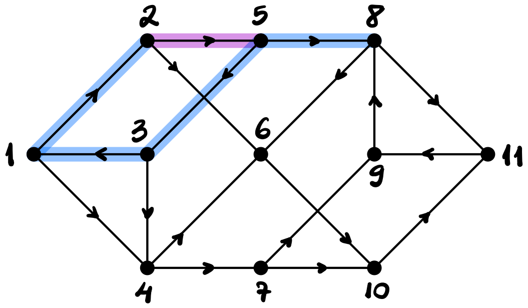

$\langle 2, 5, 3, 1, 2, 5, 8 \rangle$ is NOT a trail

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The walk W is a cycle if W is a closed trail

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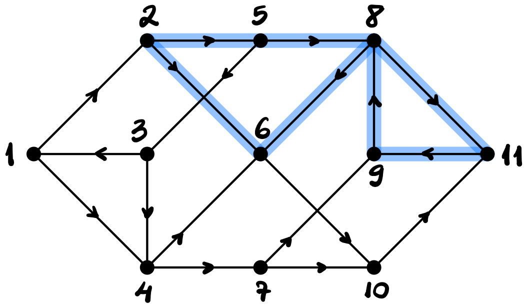

$\langle 2, 5, 8, 11, 9, 8, 6, 2 \rangle$ is a cycle

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The walk W is a cycle if W is a closed trail

This 'W' could be replaced with 'it' without losing clarity.

However, when in doubt, don't be afraid to repeat the same words/symbols.

In math, precision trumps the writing guideline to avoid repetition

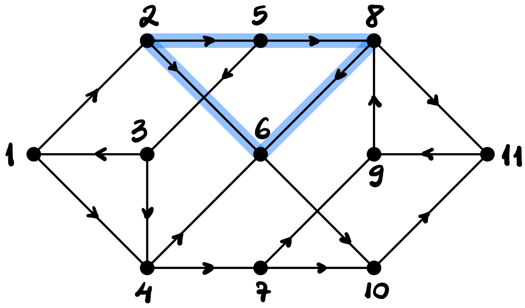
Walks

Let $D := (V, A)$ be a digraph. Let $W := \langle v_0, v_1, \ldots, v_\ell \rangle$ be a walk in D.

The walk W is a circuit if it is a cycle and

the function $i \in [\ell] \longmapsto v_i$ is injective

Example



$\langle 2, 5, 8, 6, 2 \rangle$ is a circuit

Walks

Let $D := (V, A)$ be a digraph. Let $W := \langle v_0, v_1, \dots, v_\ell \rangle$ be a walk in D.

We call W a (v_0, v_ℓ)-walk.

If W is a path, we call W a (v_0, v_ℓ)-path.

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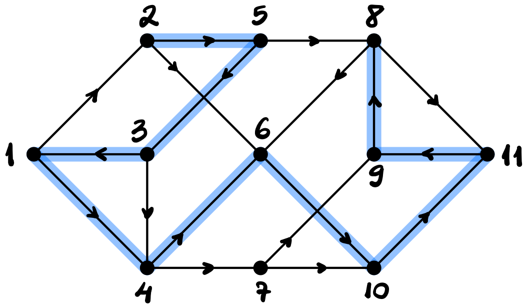

$\langle 2, 5, 3, 1, 4, 6, 10, 11, 9, 8 \rangle$ is a $(2, 8)$-path

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Let $r, s \in V$. An rs-path is shortest or minimum-length if it is an optimal solution for

Minimize ℓ

subject to $\langle v_0, v_1, \dots, v_\ell \rangle$ is an rs-path in D

The Single-Source Shortest Paths Problem

Given: a digraph $D := (V, A)$ and a vertex $r \in V$

Find: for each $v \in V$, a shortest rv-path in D

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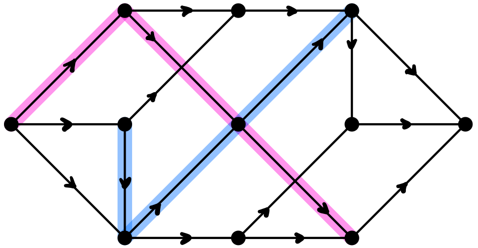
Arc-Disjoint Walks

Let $D := (V, A)$ be a digraph. Let W_1 and W_2 be walks in D.

We call W_1 and W_2 **arc-disjoint** if $A(W_1) \cap A(W_2) = \emptyset$.

Example



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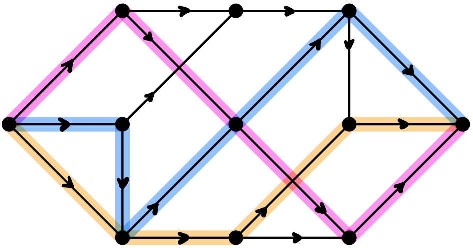

$\underbrace{\text{Packing}}$ Arc-Disjoint rs-Paths

typically refers to finding a largest collection of pairwise disjoint objects

Given: a digraph $D := (V, A)$ and distinct vertices $r, s \in V$

Find: an optimal solution for

$$\text{Maximize } |\mathcal{P}|$$

$$\text{subject to } \mathcal{P} \text{ is a set of pairwise arc-disjoint rs-paths in } D$$

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$\boxed{\text{read as "calligraphic pea" or "script pea"}}$

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$$\boxed{\begin{array}{l} \text{Maximize } k \\ \text{subject to } \{P_1, \ldots, P_k\} \text{ is a set of pairwise arc-disjoint rs-paths in } D \end{array}}$$

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ALSO, index ← singular

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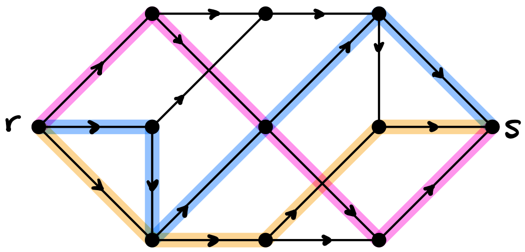

Subdigraphs & Arc Deletions

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For each $a \in A$, it is usual to write $D - \{a\}$ as $\underbrace{D - a}_{\text{called the digraph obtained from } D \text{ by deleting } a}$.

Minimum rs-Path Transversal / Cover

Given: a digraph $D:=(V,A)$ and distinct vertices $r,s \in V$

Find: an optimal solution for

$$\left\{\begin{array}{ll} \text{Minimize} & |B| \\ \text{subject to} & B \subseteq A \\ & \text{each } rs\text{-path in } D \text{ traverses some arc of } B \end{array}\right.$$

equivalently

$$\boxed{\begin{array}{ll} \text{Minimize} & |B| \\ \text{subject to} & B \subseteq A \\ & D-B \text{ has no } rs\text{-path} \end{array}}$$

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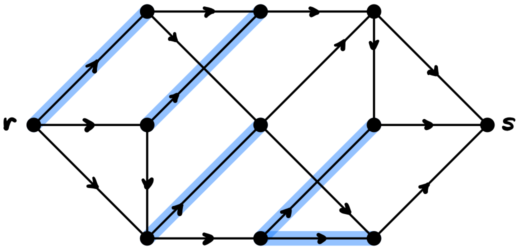

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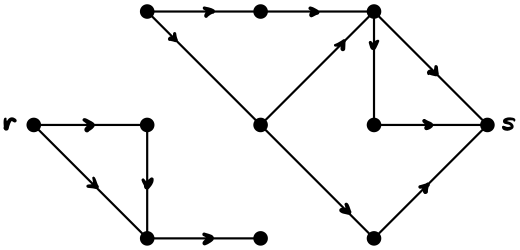

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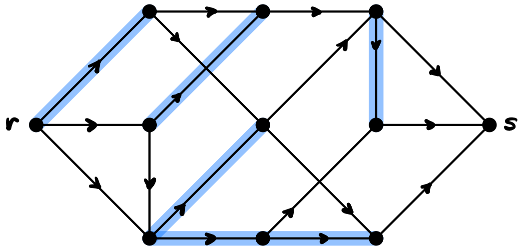

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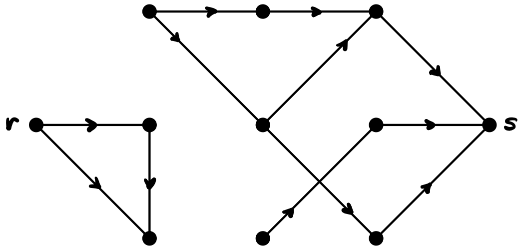

Minimum rs-Path Transversal / Cover

Given: a digraph $D := (V, A)$ and distinct vertices $r, s \in V$

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Minimize $|B|$

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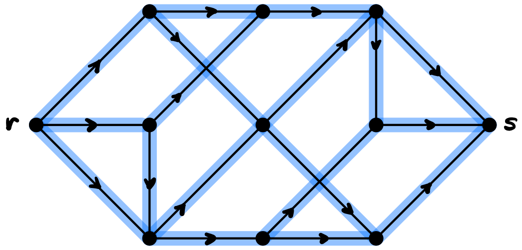

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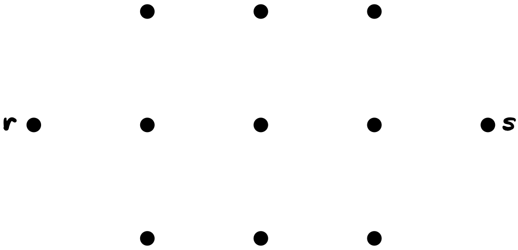

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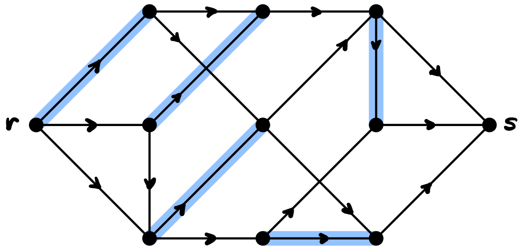

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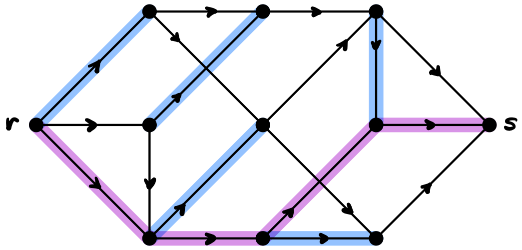
Example ... NOT!



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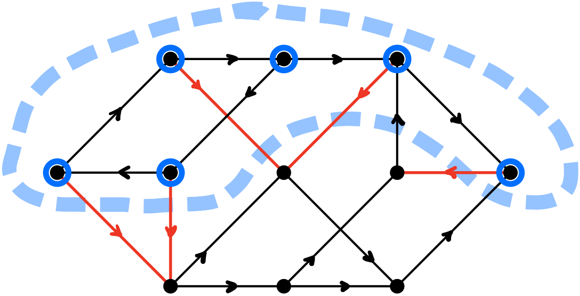
The next concepts/problems are related to this, but not in an obvious way

Cuts

Let $D := (V, A)$ be a digraph.

For each $S \subseteq V$, denote $\delta^{out}_{D}(S) := \{ uv \in A : u \in S \text{ and } v \notin S \}$

Example



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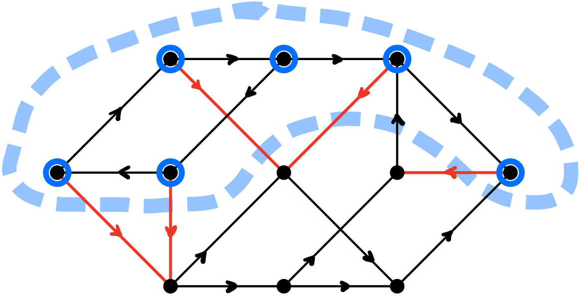

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called a shore of the cut

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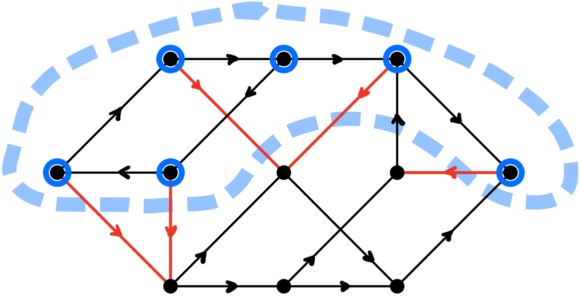

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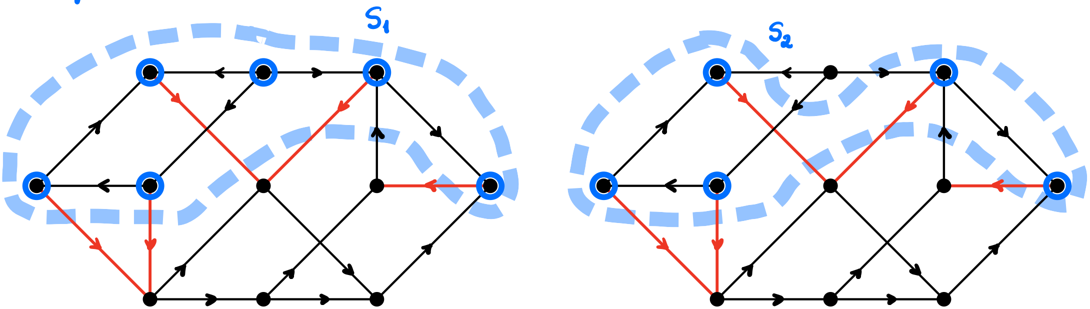

$$\delta^{out}_D(S_1) = \delta^{out}_D(S_2)$$

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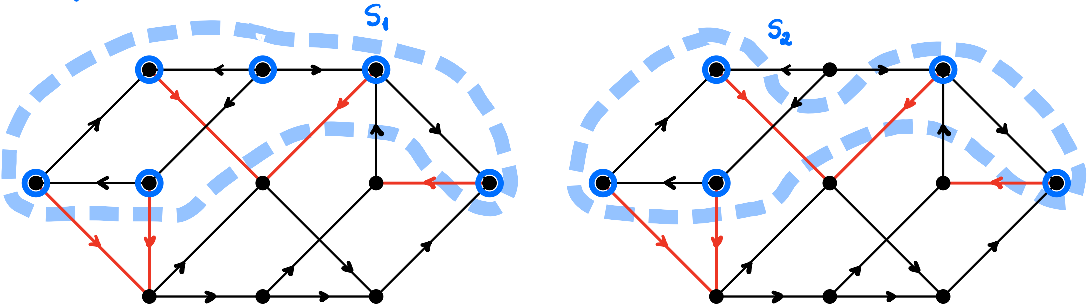

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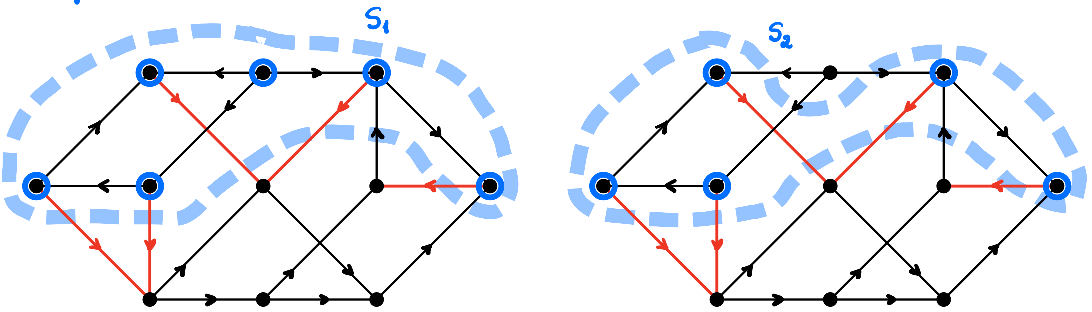

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Combinatorial Optimization

Typically, we are interested in developing (efficient) algorithms for problems of the following shape:

Given: a(n undirected) graph G or a $\overbrace{\text{directed graph}}^{\text{a digraph}}$ D

and a cost function c on the set of vertices, edges, or arcs of G or D

Find: an optimal solution for an $\underbrace{\text{optimization problem}}_{\text{and what is that?}}$

defined by G or D, and by c

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and what is that?

WATCH OUT for uniqueness or you WILL make a mistake

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the index D may be omitted if obvious from context

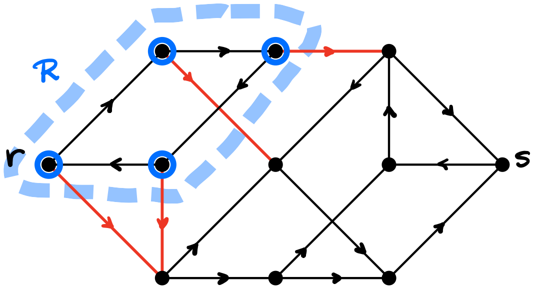
rs-Cuts

Let $D := (V,A)$ be a digraph, and let $r, s \in V$ be distinct.

A subset $B \subseteq A$ is called an rs-cut if there is $R \subseteq V$ s.t.

$r \in R \subseteq V \setminus \{s\}$ and $B = \delta^{out}(R)$

Example



Minimum rs-Cut Problem

Given: a digraph $D := (V, A)$ and distinct vertices $r, s \in V$

Find: an optimal solution for

$$\text{Minimize} \quad |\delta^{out}(R)|$$

$$\text{subject to} \quad r \in R \subseteq V \setminus \{s\}$$

Maximum Packing of rs-Cuts

Given: a digraph $D := (V, A)$ and distinct vertices $r, s \in V$

Find: an optimal solution for

the power set of V

$$
\begin{aligned}
\text{Maximize} \quad & |\mathcal{R}| \\
\text{subject to} \quad & \mathcal{R} \subseteq \boxed{\mathcal{P}(V)}, \\
& \text{for each } R \in \mathcal{R}, \text{ one has } r \in R \subseteq V \setminus \{s\}, \\
& \{\, \delta^{out}(R) \mid R \in \mathcal{R} \,\} \text{ is pairwise disjoint}
\end{aligned}
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Given: a digraph $D:=(V,A)$ and distinct vertices $r,s \in V$

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read as "calligraphic arr" or "script arr"

for each $R \in \mathcal{R}$, one has $r \in R \subseteq V \setminus \{s\}$,

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`\mathcal{R}` in LaTeX

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the power set of V

3 set-building notations

1. List the elts., e.g. $\{1,2,5,2,7\}$

2. Filter the elements of an already-proven-to-exist set A,
e.g. $\{a \in A : P(a)\}$

3. Map a function $f: A \to B$ over the elts. of A, e.g., $\{f(a) \mid a \in A\}$

Maximum Packing of rs-Cuts

Given: a digraph $D := (V, A)$ and distinct vertices $r, s \in V$

Find: an optimal solution for

$$\begin{aligned} \text{Maximize} \quad & |\mathcal{R}| \\ \text{subject to} \quad & \mathcal{R} \subseteq \boxed{\mathcal{P}(V)}, \\ & \text{for each } R \in \mathcal{R}, \text{ one has } r \in R \subseteq V \setminus \{s\}, \\ & \{\delta^{out}(R) \mid R \in \mathcal{R}\} \text{ is pairwise disjoint} \end{aligned}$$

the power set of V

3 set-building notations

1. List the elts., e.g. $\{1, 2, 5, 2, 7\}$

REQUIRED

2. Filter the elements of an already-proven-to-exist set A,
e.g. $\{a \in A : P(a)\}$

3. Map a function $f: A \to B$ over the elts. of A, e.g., $\{f(a) \mid a \in A\}$

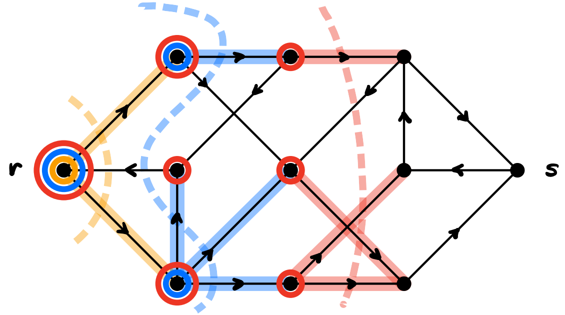
Maximum Packing of rs-Cuts – Presentation with Indices

Given: a digraph $D := (V, A)$ and distinct vertices $r, s \in V$

Find: an optimal solution for

$$\begin{aligned} &\text{Maximize} && k \\ &\text{subject to} && r \in R_i \subseteq V \setminus \{s\} \quad \text{for each } i \in [k] \\ & && \{\delta^{out}(R_1), \ldots, \delta^{out}(R_k)\} \text{ is pairwise disjoint} \end{aligned}$$

Example



Minimum Covering by Stable Sets

Given: a graph $G := (V, E)$

Find: an optimal solution for

Minimize $|\mathcal{S}|$

subject to $\mathcal{S} \subseteq \mathcal{P}(V)$

each $S \in \mathcal{S}$ is stable in G

$\bigcup \mathcal{S} = V$

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\bigcup in LaTeX for the prefix operator

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Minimize k

subject to $S_1, \dots, S_k$ are stable sets in G

$S_1 \cup \dots \cup S_k = V$

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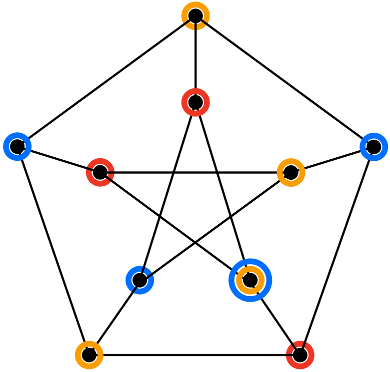

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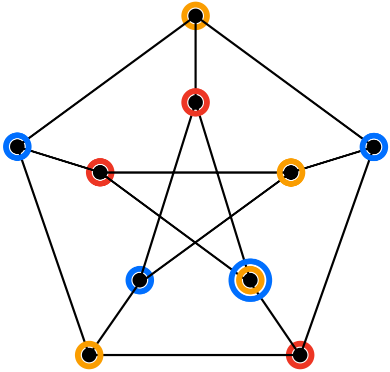

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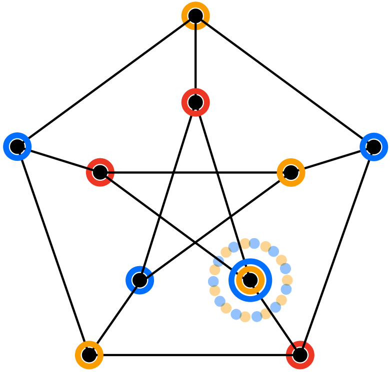
since subsets of stable sets are stable, we may demand $S_1, \dots, S_k$ to be pairwise disjoint

---

Example:



Partitions

Let S be an arbitrary set.

A partition of S is a collection $\mathcal{P}$ of subsets of S s.t.

(i) each $P \in \mathcal{P}$ is nonempty,

(ii) $\mathcal{P}$ is pairwise disjoint, and

(iii) $\bigcup \mathcal{P} = S$.

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Examples:

$\{\{3,5,7,9\},\{1\},\{2,4,6,8,10\}\}$ is a partition of $\{3,5,7,9,1,2,4,6,8,10\}$

$\{[x,x+1) \mid x \in \mathbb{Z}\}$ is a partition of $\mathbb{R}$

$\emptyset$ is a partition of $\emptyset$

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NEVER EVER call $P \in \mathcal{P}$ a partition of $\mathcal{P}$

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$\{[x,x+1) \mid x \in \mathbb{Z}\}$ is a partition of $\mathbb{R}$

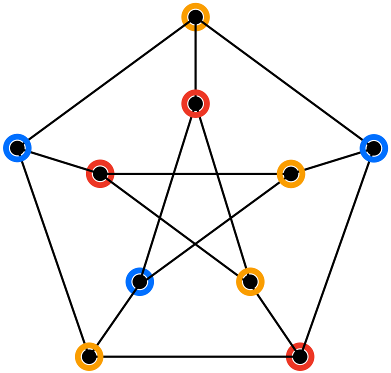
$\varnothing$ is a partition of $\varnothing$

Graph Colorings

Let $G := (V,E)$ be a graph.

A coloring of G is a partition $\mathcal{P}$ of V s.t. each $P \in \mathcal{P}$ is stable in G.

Example:



Minimum Coloring Problem

Given: a graph $G := (V, E)$

Find: an optimal solution for

Minimize $|S|$

subject to S is a coloring of G

Walks in Graphs

Let $G := (V, E)$ be a graph. A sequence $W := \langle v_0, v_1, \dots, v_\ell \rangle$ of vertices of G is a **walk** if, for each $i \in [\ell]$, one has $v_{i-1}v_i \in E$.

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$\ell \in \mathbb{N}$ is called the **length** of the walk

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Denote $V(W) := \{ v_0, v_1, \dots, v_\ell \}$ and $E(W) := \{ v_{i-1}v_i \mid i \in [\ell] \} \subseteq E$

We say that W is a walk from v_0 to v_ℓ or a (v_0, v_ℓ)-walk.

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We say that W is a walk from v_0 to v_ℓ or a (v_0, v_ℓ)-walk.

The walk W is:

(i) **closed** if $\ell \geq 1$ and $v_0 = v_\ell$;

(ii) a **trail** if the function $i \in [\ell] \mapsto v_{i-1}v_i \in E$ is injective;

(iii) a **path** if the function $i \in \{0, \dots, \ell\} \mapsto v_i$ is injective;

(iv) a **circuit** if W is a closed trail and $i \in [\ell] \mapsto v_i$ is injective.

Forests

A graph is a forest if it has no circuits.

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Let $G := (V, E)$ be a graph. We call a subset $F \subseteq E$ a **forest** (of G) if (V, F) is a forest.

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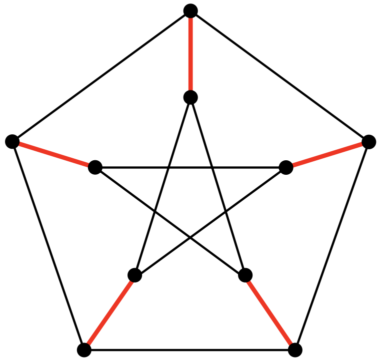

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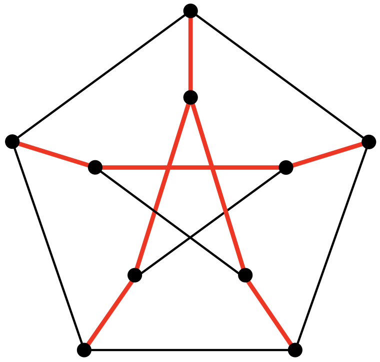

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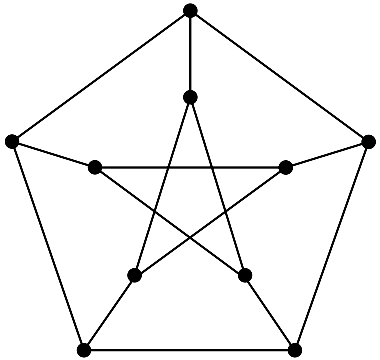

Covering by Forests

Given: a graph $G := (V, E)$

Find: an optimal solution for

Minimize $|\mathcal{F}|$

subject to $\mathcal{F} \subseteq \mathcal{P}(E)$

each $F \in \mathcal{F}$ is a forest of G

$\bigcup \mathcal{F} = E$

Covering by Forests

Given: a graph $G := (V, E)$

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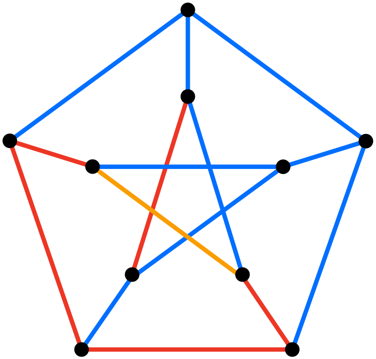
Example:



Connected Graphs & Trees

Let $G := (V, E)$ be a graph.

We call G **connected** if, for each $u, v \in V$, there is a uv-walk in G.

If G is not connected, we call G **disconnected**.

If G is connected and a forest, we call G a **tree**.

Subgraphs

Let $G := (V, E)$ and $G' := (V', E')$ be graphs.

We say that G' is a subgraph of G if $V' \subseteq V$ and $E' \subseteq E$.

If $G' \neq G$, we call G' a proper subgraph of G.

If $V' = V$, we call G' a spanning subgraph of G.

Subgraphs & Spanning Trees

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If $V' = V$, we call G' a spanning subgraph of G.

We call a subset $F \subseteq E$ a spanning tree (of G) if (V,F) is a tree.

Tree Packing

Given: a graph $G := (V, E)$

Find: an optimal solution for

$$\begin{aligned} \text{Maximize} \quad & |\mathcal{T}| \\ \text{subject to} \quad & \mathcal{T} \subseteq \mathcal{P}(E), \\ & \text{each } T \in \mathcal{T} \text{ is a spanning tree of } G \\ & \mathcal{T} \text{ is pairwise disjoint} \end{aligned}$$

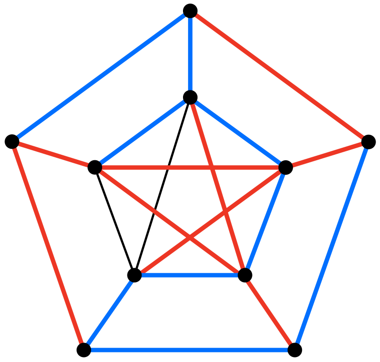
Tree Packing – Presentation with Indices

Given: a graph $G := (V, E)$

Find: an optimal solution for

$$\begin{array}{lll} \text{Maximize} & k & \\ \text{subject to} & T_i \subseteq E \text{ is a spanning tree of } G & \text{for each } i \in [k] \\ & \{T_1, \ldots, T_k\} \text{ is pairwise disjoint} & \end{array}$$

Example



In-Memory Digraph Representation

Disclaimers

(i) in this course, we are NOT interested in the fastest known algorithms and data structures

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> except when dealing with verification algorithms

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also by Cobham

introduced by Jack Edmonds in 1965 in the context of combinatorial optimization

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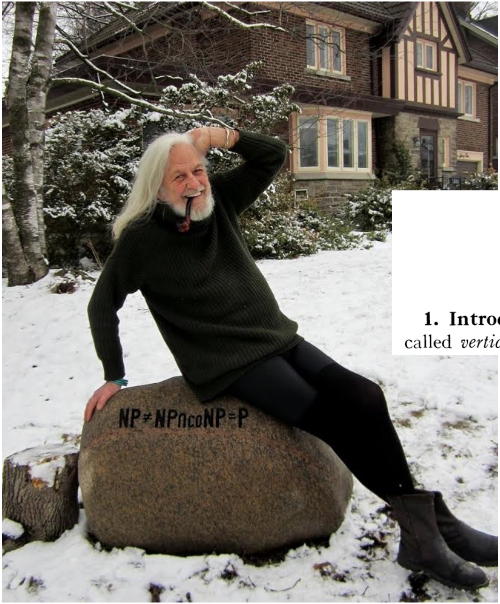
Jack Edmonds



Edmonds at his Waterloo home

PATHS, TREES, AND FLOWERS

JACK EDMONDS

1. Introduction. A *graph G* for purposes here is a finite set of elements called *vertices* and a finite set of elements called *edges* such that each edge

Jack Edmonds

From "Paths, Trees, and Flowers", 1965

"For practical purposes computational details are vital. However, my purpose is only to show as attractively as I can that there is an efficient algorithm. According to the dictionary, "efficient" means "adequate in operation or performance." This is roughly the meaning I want—in the sense that it is conceivable for maximum matching to have no efficient algorithm. Perhaps a better word is "good."

I am claiming, as a mathematical result, the existence of a *good* algorithm for finding a maximum size matching in a graph.

There is an obvious finite algorithm, but that algorithm increases in difficulty exponentially with the size of the graph. It is by no means obvious whether *or not* there exists an algorithm whose difficulty increases only algebraically with the size of the graph."

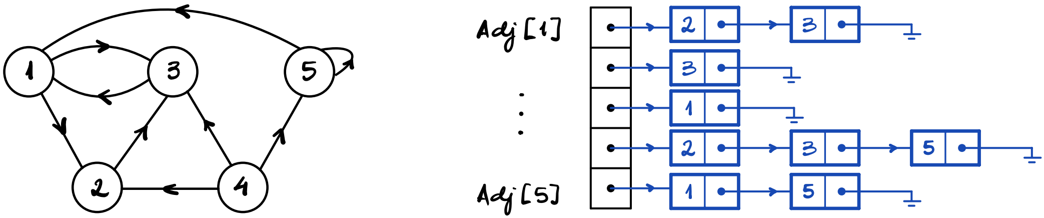
polynomially

Jack Edmonds

From "Optimum branchings," 1967

"I conjecture that there is no good algorithm for the traveling salesman problem. My reasons are the same as for any mathematical conjecture: (1) It is a legitimate mathematical possibility, and (2) I do not know."

Typical Representation with Adjacency Lists



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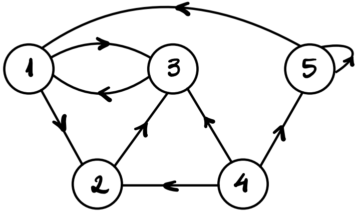
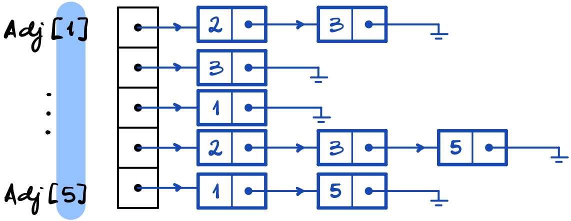
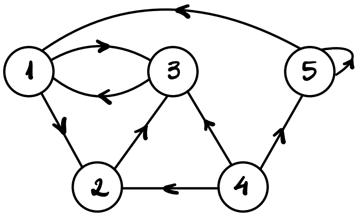


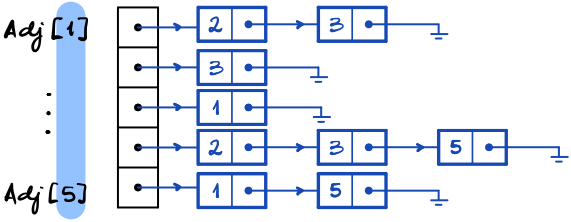


vertices are 1, ..., n

(or 0, ..., n−1)

Typical Representation with Adjacency Lists





vertices are $1, \ldots, n$

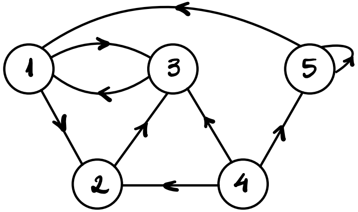
(or $0, \ldots, n-1$)

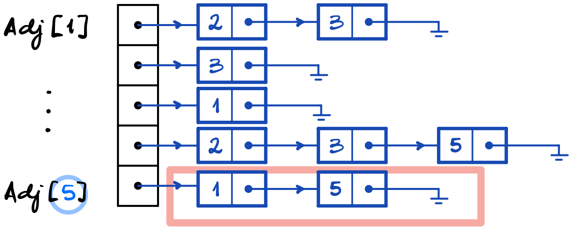
later we'll use

constant-time indexing/random access

by each $v \in V$, whatever V is

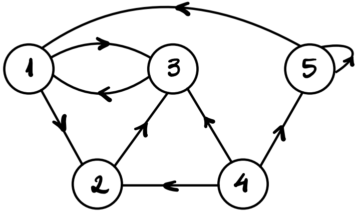
Typical Representation with Adjacency Lists

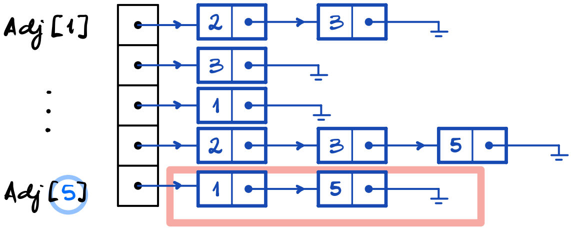




linked list of arcs which have 5 as tail

Typical Representation with Adjacency Lists

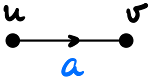




linked list of arcs which have 5 as tail

Terminology: if $a := uv$ is an arc of a digraph,

we call v the head of a and u the tail of a



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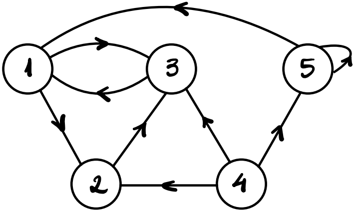
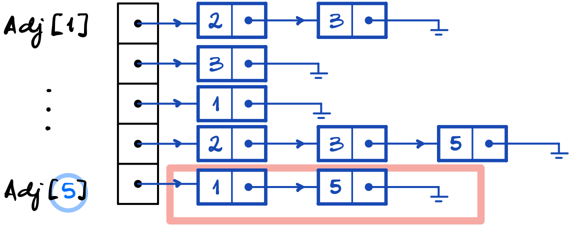
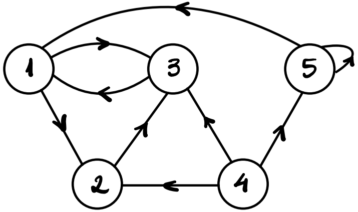


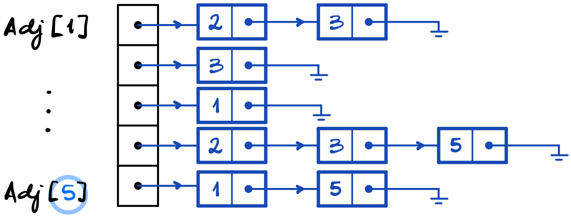


linked list of arcs which have 5 as tail

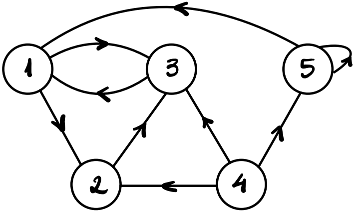
NOT crucial

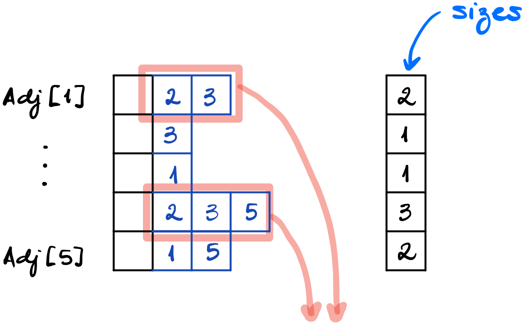
Typical Representation with Adjacency Lists





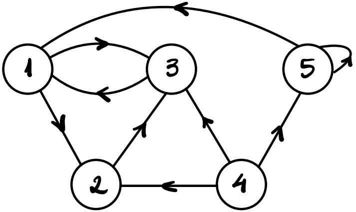
~~Typical~~ Representation with Adjacency ~~Lists~~ Arrays

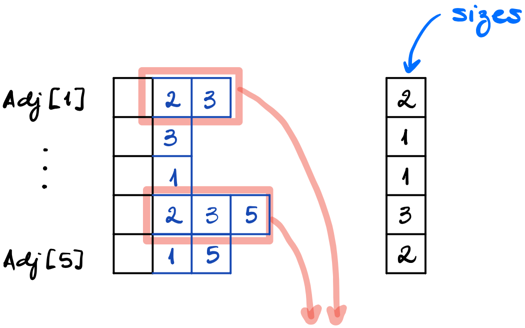




arrays with (possibly) different sizes

~~Typical~~ Representation with Adjacency ~~Lists~~ Arrays





arrays with (possibly) different sizes

constant-time indexing/random access of arcs is clumsy/awkward

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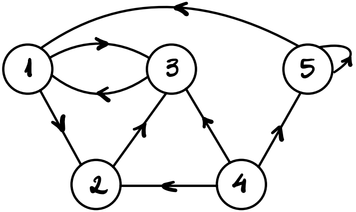
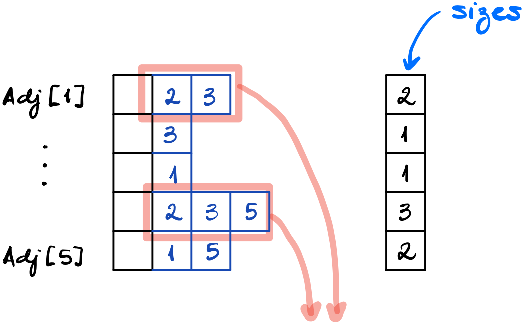




arrays with (possibly) different sizes

constant-time indexing/random access of arcs is clumsy/awkward

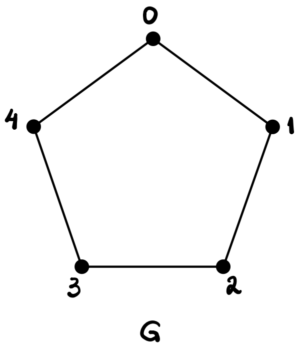
where to store extra "attributes" for each vertex/arc?

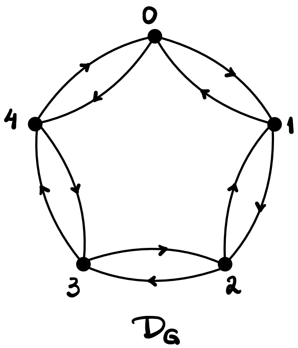
Graph Representation

A graph $G = (V, E)$ is usually represented as its symmetric digraph

$$D_G := \left(V, \bigcup_{uv \in E} \{(u,v),(v,u)\}\right).$$

Example





Graph Representation

A graph $G=(V,E)$ is usually represented as its **symmetric digraph**

$$D_G := \Big(V, \bigcup_{uv\in E} \{(u,v),(v,u)\}\Big).$$

syntactic sugar for $\bigcup\{\{(u,v),(v,u)\} : uv \in E\}$

Example